

Bayer superpixel weight derivation

Thorlabs LP126CU camera under Olympus U-LH100IR illumination

InterferoLab

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Contents

1	Motivation	1
2	Optimal weighting criterion	2
3	Spectral model of the lamp	2
4	Input data	2
5	Computation procedure	3
6	Results	3
6.1	Comparison with 2800 K	3
7	Use of the weights in the system	3
8	Limitations and possible improvements	4

1. Motivation

The Thorlabs LP126CU is a color CMOS sensor with a **Bayer color filter array** (CFA) in **BGGR** pattern. Each physical pixel measures only one of the three colors, according to its position in the 2×2 grid:

B	G ₁
G ₂	R

*Corrected in September 2026: this report previously said RGGB and drew the block with R at the top left. The sensor reports `filter_array_phase = "blue"`, i.e. **BGGR**, and the application now reads that phase from the camera instead of assuming it. The weights derived below are unaffected, because they depend only on the count of photosites per block (one red, two green, one blue) and not on their arrangement; what the pattern fixes is which photosite of the block receives w_R and which receives w_B .*

Coherence scanning interferometry (CSI) works with a *monochrome* intensity signal: the gray value of each position (y, x) as a function of the axial position z . To exploit the signal of all colors without resorting to demosaicing (which correlates neighbouring pixels at the CFA period and produces a measurable spike at 0.5 cycles/pixel in the reconstructed height maps; see the *Camera characterization report*, whose earlier attribution of the second fringe harmonic to demosaicing has

been withdrawn), each 2×2 block is collapsed into a single **superpixel**:

$$I_{\text{sp}} = w_R R + w_{G_1} G_1 + w_{G_2} G_2 + w_B B. \quad (1)$$

The optimal choice of the weights w_c maximizes the signal-to-noise ratio (SNR) of the superpixel.

2. Optimal weighting criterion

The CSI interference signal is proportional to the received light intensity. For each channel $c \in \{R, G_1, G_2, B\}$, the captured signal is:

$$S_c = \int_{\lambda} S_{\text{lamp}}(\lambda) T_{\text{IR}}(\lambda) \text{QE}_c(\lambda) d\lambda \quad (2)$$

where:

- $S_{\text{lamp}}(\lambda)$ — emission spectrum of the lamp (spectral power density, a.u.).
- $T_{\text{IR}}(\lambda)$ — transmission of the IR filter built into the camera.
- $\text{QE}_c(\lambda)$ — quantum efficiency of channel c (taken from the Thorlabs datasheet, without the IR filter, which is applied separately).

The optimal weight in the maximum-SNR sense is simply proportional to the received signal:

$$w_c \propto S_c, \quad \sum_c w_c = 1. \quad (3)$$

The Bayer block contains *two* green pixels for every red and blue one, so the normalization denominator is $S_R + 2S_G + S_B$, not $S_R + S_G + S_B$:

$$w_R = \frac{S_R}{S_R + 2S_G + S_B}, \quad w_{G_1} = w_{G_2} = \frac{S_G}{S_R + 2S_G + S_B}, \quad w_B = \frac{S_B}{S_R + 2S_G + S_B}. \quad (4)$$

3. Spectral model of the lamp

The lamp in use is an **Olympus U-LH100IR** (100 W, 12 V), a tungsten halogen lamp. Halogen lamps operate at higher filament temperatures (~ 3200 K) than conventional incandescent ones (~ 2800 K) thanks to the halogen cycle that regenerates the filament.

The emission spectrum is approximated by **Planck's law** for a black body at $T = 3200$ K:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{e^{hc/(\lambda k_B T)} - 1} \quad (5)$$

Only the spectral shape matters, not the absolute magnitude.

Note: If the measured spectral curve of the U-LH100IR is available, $B(\lambda, T)$ should be replaced by that curve for better accuracy.

4. Input data

The script `compute_bayer_weights.py` reads four CSV files extracted from the plots in the Thorlabs datasheet with the WebPlotDigitizer tool:

File	Contents
<code>ir_filter_transmission.csv</code>	$T_{\text{IR}}(\lambda)$ in %
<code>qe_red.csv</code>	$\text{QE}_R(\lambda)$ in %
<code>qe_green.csv</code>	$\text{QE}_G(\lambda)$ in %
<code>qe_blue.csv</code>	$\text{QE}_B(\lambda)$ in %

All data are interpolated onto a common 400–1000 nm grid with a 1 nm step. Values outside the measured range are treated as zero (null sensor response), which is the correct conservative assumption.

5. Computation procedure

1. **Load and interpolate** the four curves onto the grid $\lambda \in [400, 1000]$ nm.
2. **Compute the lamp spectrum** $B(\lambda, 3200)$ from Planck’s law.
3. **Compute the integrand** per channel:

$$f_c(\lambda) = B(\lambda, 3200) \cdot T_{\text{IR}}(\lambda) \cdot \text{QE}_c(\lambda) \quad (6)$$

4. **Integrate numerically** with the trapezoidal rule: $S_c = \int f_c(\lambda) d\lambda$.
5. **Normalize** according to equation (4): denominator = $S_R + 2S_G + S_B$.

The script also generates a verification figure with three panels: (a) the interpolated input curves, (b) the normalized lamp spectrum, and (c) the integrands $f_c(\lambda)$ normalized by their own maximum.

6. Results

With $T = 3200$ K (Olympus U-LH100IR) and the LP126CU curves:

Channel	Weight	Integrated signal (a.u.)
w_R	0.257090	S_R
w_{G_1}	0.311272	S_G
w_{G_2}	0.311272	S_G
w_B	0.120366	S_B
Sum	1.000000	

The green channel receives the largest individual weight because its QE is high across the visible range and the halogen lamp emits strongly in that region. The blue channel is the weakest: the QE is low (< 20%) in the blue, and the lamp spectrum decays exponentially towards short wavelengths.

6.1. Comparison with 2800 K

T (K)	w_R	$w_{G_1} = w_{G_2}$	w_B	Change in w_B
2800	0.2829	0.3073	0.1026	—
3200	0.2571	0.3113	0.1204	+17%

The correct temperature (3200 K vs. 2800 K) increases the blue channel’s weight by 17% and reduces the red one’s by 9%. Using 2800 K would over-weight red (which carries less real signal from the halogen lamp) and under-weight blue.

7. Use of the weights in the system

Since the July 2026 reorganization, the **single source** of the weights on the Python side is the module `utils/camera_constants.py`. It is consumed by:

1. **Acquisition** (`backend/acquisition/acquisition_controller.py`): the function `_bayer_to_superpixel` which combines the four sub-pixels of the raw Bayer block in real time during acquisition.

2. **Offline processing** (`scripts/process_superpixel.py`): conversion of already-acquired datasets to the superpixel format.
3. **GUI**: the default values of the weight spinboxes in the processing panel.

The C++ **analysis** keeps a copy of these values in `backend/analysis/include/config.hpp` (constants `GRAY_WEIGHT_R/G/B`), which must be kept *synchronized by hand* with `utils/camera_constants.py` on recalibration (a regression test detects divergence). In that context the images are already demosaiced and the G channel represents the mean of G_1 and G_2 , so its weight is doubled:

$$W_G^{\text{RGB}} = w_{G_1} + w_{G_2} = 0.622544. \quad (7)$$

8. Limitations and possible improvements

- **Curves extracted from plots**: the CSVs come from WebPlotDigitizer applied to the datasheet plots, not from tabulated data. The digitization error is estimated at $\pm 1\text{--}2\%$ per channel.
- **Lamp spectrum**: a black body at 3200 K is used as an approximation. The real color temperature of the U-LH100IR can vary slightly with the supply voltage. A measured spectral curve would improve accuracy.
- **Optical-path transmission**: the model does not include the spectral transmission of the interferometric objective, the beam splitter, or the condenser of the Olympus microscope. These elements can shift the optimal weights non-negligibly (the experimental coherence length suggests a spectral narrowing of a factor ~ 2 with respect to the camera-only model).